

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB
(Common to II year – AERO, MECH)

II B.TECH II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE61	PCC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

1. To introduce the concepts of electrical circuits and its components
2. To study and understand the different types of DC/AC machines and Transformers.
3. To introduce the concepts of diodes & transistors, and
4. To impart the knowledge of various transistors configurations, characteristics and applications.

Course Outcomes:

LIST OF EXPERIMENTS

PART A : ELECTRICAL

1	a. Identification, Specifications, Testing of R, L, C Components (Color Codes), Bread Boards, Diodes, BJTs. b. Verification of Ohms law	
2	Verification of KCL and KVL.	
3	Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)	
4	Swinburne's test on dc shunt motor (Predetermination of efficiencies)	
5	OC and SC tests on Single-phase transformer.	
6	Brake test on 3-phase Induction motor.	
	PART B : ELECTRONICS	
7	Study and operation of Function Generator Regulated Power Supply (RPS) CRO.	
8	PN Junction Diode Characteristics (Forward bias, Reverse bias)	
9	Zener Diode Characteristics	
10	Rectifier without Filters (Full wave & half wave).	
11	Rectifier with Filters (Full wave & half wave).	
12	Transistor CE Characteristics (Input and Output)	

Text Books:

1. Basic Electrical Engineering by *M.S.Naidu and S.Kamakshaiah* TMH
2. Electronic Devices and circuits by *J.Millman, C.C.Halkias and Satyabrata Jit* 2ed.,

Reference Books:

1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, “Basic Electrical, Electronics and Computer Engineering”, Tata McGraw Hill, Second Edition, (2006).
2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press (2005).
3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd, (1994).
4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, (2002).